

Skeletal system

Some of the hardest parts of the body make up the skeleton. Bone contains living cells but is also packed with hard minerals. This helps it support the stresses and strains of the moving body, and to protect soft organs, too.

Circulatory system

No living cell is very far from a blood vessel. The circulatory system serves as a lifeline to every cell. It circulates food, oxygen, and chemical triggers—such as hormones—as well as transporting waste to the excretory organs.

Digestive system

By processing incoming food, the digestive system is the source of fuel and nourishment for the entire body. It breaks down food to release nutrients, which then seep into the bloodstream to be circulated to all living cells.

Muscular system

The moving parts of the body rely on muscles that contract when triggered to do so by a nerve impulse or chemical trigger. Contraction shortens the muscle, which pulls on a part of the body to cause motion.

The hard casing of the skull protects the brain.

The heart pumps blood around the body.

The muscles in the center of the chest assist with the movements of breathing.

Adult humans have a total of **206 bones** in their skeletal system.

Body systems

A living body has so many working parts that cells, tissues, and organs will only run smoothly by cooperating with one another in a series of highly organized systems.

Each system is designed to carry out a particular function essential to life—whether breathing, eating, or reproducing. Just as organs are interconnected in organ systems, the systems interact, and some organs, such as the pancreas, even belong to more than one system.

Blood vessels spread all the way to the extremities of the body.

The coiled-up small intestine has a large surface for absorbing nutrients.

Human body systems

There are 12 systems of the human body, of which 8 of the most vital are shown here. The others are the urinary system (see pp.162–163), the integumentary system (skin, hair, and nails), the lymphatic system (which drains excess fluid), and the endocrine system (which produces hormones).